

Visitor Management Backend - Technical Documentation

Spring Boot 4.1.0 · Java 21 · PostgreSQL 17 · Hibernate 7.4

1. Architecture Overview

```

+-----+
|                                     |
|             REST Controllers        |
| TimeSlotController · VisitorController |
| RegistrationController · BriefingController |
| EventController · FeedbackController |
| EducationalProgramController · BookingController |
+-----+
|                                     |
|             Services                |
| SchedulingServiceImpl · RegistrationServiceImpl |
| EventServiceImpl · FeedbackServiceImpl |
| EducationalProgramServiceImpl · BookingServiceImpl |
| BookingReminderJob (rappel 24h) |
| SchedulingRules (constants) |
+-----+
|                                     |
|             Repositories (JPA)      |
| TimeSlotRepository · VisitorRepository |
| RegistrationRepository · BriefingRepository |
| EventRepository · FeedbackRepository |
| SurveySendRepository · TourStopRepository |
| WorkshopRepository · AgriActivityRepository |
| BookingRepository |
+-----+
|                                     |
|             JPA Entities            |
| TimeSlot · Visitor · Registration · Briefing · Event |
| Feedback · SurveySend · TourStop · Workshop |
| AgriActivity · Booking |
| (extend BaseEntity: id + createdAt + updatedAt) |
+-----+
|                                     |
|             core/ - shared infrastructure |
| BaseEntity · GlobalExceptionHandler · ApiError |
| ResourceNotFoundException · ConflictException |
| BusinessException · NotificationService (console/SMTP) |
+-----+

```

Packages:

- core/ - shared infra only (BaseEntity, exceptions, handler, notification)
- modules/visitormanagement/ - Scheduling + Registration sub-modules
- Enums live in the module package (not core/)

2. Database Schema

Entity-Relationship Diagram

+-----+ +-----+			
time_slot		visitor	
+-----+ +-----+			
id	BIGINT PK	id	BIGINT PK
slot_date	DATE NOT NULL	full_name	VARCHAR(150) NOT NULL
start_time	TIME NOT NULL	group_size	INT NOT NULL, DEFAULT 1
end_time	TIME NOT NULL	email	VARCHAR(200) UNIQUE
max_capacity	INT NOT NULL	phone	VARCHAR(30)
status	VARCHAR(20) NOT NULL	language	VARCHAR(40)
guide_id	BIGINT	visitor_type	VARCHAR(20) NOT NULL
created_at	INSTANT NOT NULL	special_needs	VARCHAR(500)

```

| updated_at INSTANT | NOT NULL | created_at INSTANT | NOT NULL
+-----+-----+
| 1 | 1 | |
| 1 | 1 |
| * | 1 | * |
+-----+-----+
| registration |
+-----+-----+
| id BIGINT | PK |
| visitor_id BIGINT | FK -> visitor (NOT NULL) |
| time_slot_id BIGINT | FK -> time_slot (nullable, NULL for |
| | event-only registrations) |
| event_id BIGINT |
| visit_purpose VARCHAR(20) | NOT NULL |
| is_prospect BOOLEAN | NOT NULL, DEFAULT false |
| status VARCHAR(20) | NOT NULL, DEFAULT 'PENDING' |
| created_at INSTANT | NOT NULL |
| updated_at INSTANT | NOT NULL |
| UNIQUE (visitor_id, time_slot_id) |
+-----+-----+
| 1 |
| 1 |
| 1 |
+-----+-----+
| briefing |
+-----+-----+
| id BIGINT | PK |
| registration_id | FK -> registration |
| | (NOT NULL, UNIQUE) |
| delivered_at INSTANT |
| staff_member VARCHAR(150) |
| signature VARCHAR(500) |
| status VARCHAR(20) | NOT NULL, 'PENDING' |
| created_at INSTANT | NOT NULL |
| updated_at INSTANT | NOT NULL |
+-----+-----+

```

registration.event_id -> event.id (no FK constraint, soft link)

```

+-----+-----+
| event |
+-----+-----+
| id BIGINT | PK |
| title VARCHAR(150) | NOT NULL |
| event_type VARCHAR(30) | NOT NULL |
| start_date_time TIMESTAMP | NOT NULL |
| end_date_time TIMESTAMP | NOT NULL |
| max_capacity INT | NOT NULL, DEFAULT 10 |
| status VARCHAR(20) | NOT NULL, 'DRAFT' |
| description VARCHAR(1000) |
| location VARCHAR(150) |
| created_at INSTANT | NOT NULL |
| updated_at INSTANT | NOT NULL |
+-----+-----+

+-----+-----+
| feedback |
+-----+-----+
| id BIGINT | PK |
| visitor_id BIGINT | FK -> visitor (NOT NULL) |
| survey_send_id BIGINT | FK -> survey_send (nullable) |
| origin VARCHAR(20) | NOT NULL |

```

```

| rating          INT| NOT NULL (1..5) |
| briefing_clear  VARCHAR(1000)|
| educational_value VARCHAR(1000)|
| recommend       VARCHAR(1000)|
| comment         VARCHAR(1000)|
| routed_to      VARCHAR(60)|
| submitted_at    INSTANT| NOT NULL
| created_at      INSTANT| NOT NULL
| updated_at      INSTANT| NOT NULL
+-----+

+-----+
|          survey_send          |
+-----+

| id          BIGINT| PK
| visitor_id   BIGINT| FK -> visitor (NOT NULL)|
| channel      VARCHAR(20)| NOT NULL
| message_template VARCHAR(500)|
| sent_at      INSTANT| NOT NULL
| status       VARCHAR(20)| NOT NULL, 'SENT'|
| created_at    INSTANT| NOT NULL
| updated_at    INSTANT| NOT NULL
+-----+

+-----+
|          tour_stop          |
+-----+

| id          BIGINT| PK
| name        VARCHAR(150)| NOT NULL
| position     INT| NOT NULL
| description  VARCHAR(2000)|
| duration_minutes INT| NOT NULL
| max_capacity INT|
| location     VARCHAR(150)|
| demo        VARCHAR(1000)|
| safety_notes VARCHAR(1000)|
| active       BOOLEAN| NOT NULL, true
| created_at    INSTANT| NOT NULL
| updated_at    INSTANT| NOT NULL
+-----+

+-----+
|          workshop          |
+-----+

| id          BIGINT| PK
| name        VARCHAR(150)| NOT NULL
| duration_minutes INT| NOT NULL
| target_group VARCHAR(60)| NOT NULL
| facilitator  VARCHAR(100)|
| description  VARCHAR(1000)|
| status       VARCHAR(20)| NOT NULL, 'DRAFT'|
| created_at    INSTANT| NOT NULL
| updated_at    INSTANT| NOT NULL
+-----+

+-----+
|          agri_activity          |
+-----+

| id          BIGINT| PK
| name        VARCHAR(150)| NOT NULL
| price DECIMAL(12,2)| NOT NULL
| capacity     INT| NOT NULL
| duration_minutes INT| NOT NULL

```

```

| description VARCHAR(1000)|
| active      BOOLEAN| NOT NULL, true |
| created_at   INSTANT| NOT NULL      |
| updated_at   INSTANT| NOT NULL      |
+-----+

+-----+
|          booking          |
+-----+
| id          BIGINT| PK          |
| agri_activity_id BIGINT| FK NOT NULL|
| time_slot_id  BIGINT| FK NOT NULL|
| visitor_full_name VARCHAR(150)| NN|
| visitor_email  VARCHAR(255)| NN  |
| visitor_phone  VARCHAR(30)|      |
| people_count   INT| NOT NULL   |
| payment_method VARCHAR(20)|      |
| payment_status VARCHAR(20)| UNPAID |
| status         VARCHAR(20)| PENDING |
| reminder_scheduled_at INSTANT|      |
| confirmation_sent_at INSTANT|      |
| reminder_sent_at INSTANT|      |
| created_at     INSTANT| NOT NULL  |
| updated_at     INSTANT| NOT NULL  |
+-----+

```

Table Summary

Table	Rows (design)	Purpose
`time_slot`	~500/year	Farm tour calendar slots (2/day max)
`visitor`	~2,000	Visitor database (individuals/groups)
`registration`	~4,000	Links visitor -> time slot **or** event
`briefing`	~3,000	Safety briefing per confirmed registration
`event`	~50/year	Open days, buyer visits, school/community days
`feedback`	~3,000	Satisfaction responses (tablet + survey links)
`survey_send`	~8,000	Post-visit surveys sent by email/SMS/WhatsApp
`tour_stop`	~8	Standard guided-tour stops (seeded via API)
`workshop`	~10	Tour templates / workshops (Active / Draft / Inactive)
`agri_activity`	~5	Paid agritourism offers (catalogue)
`booking`	~2,000/yr	Paid reservations linked to an activity + time slot

Enums

Enum	Values	Table Column
`TimeSlotStatus`	`AVAILABLE`, `RESERVED`, `FULL`, `CANCELLED`, `COMPLETED`	`time_slot.status`
`RegistrationStatus`	`PENDING`, `CONFIRMED`, `REJECTED`, `CHECKED_IN`, `CANCELLED`	`registration.status`
`BriefingStatus`	`PENDING`, `DONE`	`briefing.status`
`VisitorType`	`INDIVIDUAL`, `GROUP`, `SCHOOL`, `PARTNER`	`visitor.visitor_type`
`VisitPurpose`	`TOURISM`, `PURCHASE`, `PARTNERSHIP`, `INVESTMENT`, `EDUCATION`, `OTHER`	`registration.visit_purpose`
`EventType`	`OPEN_DAY`, `PARTNER_BUYER`, `SCHOOL`, `COMMUNITY`, `OTHER`	`event.event_type`
`EventStatus`	`DRAFT`, `PUBLISHED`, `CANCELLED`, `COMPLETED`	`event.status`
`FeedbackChannel`	`ON_SITE`, `EMAIL`, `SMS`, `WHATSAPP`	`feedback.origin`
`survey_send.channel`		
`SurveyStatus`	`SENT`, `RECEIVED`	`survey_send.status`
`WorkshopStatus`	`DRAFT`, `ACTIVE`, `INACTIVE`	`workshop.status`
`BookingStatus`	`PENDING`, `CONFIRMED`, `COMPLETED`, `CANCELLED`	`booking.status`
`BookingPaymentStatus`	`UNPAID`, `PAID`, `REFUNDED`	`booking.payment_status`
`BookingPaymentMethod`	`CASH`, `ORANGE_MONEY`, `MOOV_MONEY`	`booking.payment_method`

3. REST API Endpoints

Base URL: <http://localhost:8080/api/v1>

3.1 Time Slots - [/api/v1/time-slots](#)

Method	Path	Description	Status Codes
GET	/time-slots	List all slots (optional `?date=YYYY-MM-DD`)	200
GET	/time-slots/{id}	Get slot by ID	200, 404
POST	/time-slots	Create a new slot	201, 400, 409, 422
PUT	/time-slots/{id}	Update slot details	200, 404, 422
DELETE	/time-slots/{id}	Cancel a slot (soft-delete)	204, 404, 422
POST	/time-slots/{id}/assign-guide	Assign a guide (`?guideId=N`)	200, 404, 422
GET	/time-slots/availability?date=...	Availability view for a date	200

```
{
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "guideId": 5
}
```

Create/Update Request Body:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "booked": 3,
  "status": "RESERVED",
  "guideId": 5,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

TimeSlotResponse:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "remaining": 7,
  "status": "RESERVED",
  "guideId": 5
}
```

AvailabilityResponse:

3.2 Visitors - [/api/v1/visitors](#)

Method	Path	Description	Status Codes
GET	/visitors	List all visitors	200
GET	/visitors/{id}	Get visitor by ID	200, 404
POST	/visitors	Create visitor	201, 400, 409
PUT	/visitors/{id}	Update visitor	200, 404, 409

```
{
  "fullName": "Alice Dupont",
  "groupSize": 2,
  "email": "alice@example.com",
  "phone": "+241 06 12 34 56",
}
```

```

"language": "French",
"type": "INDIVIDUAL",
"specialNeeds": "Wheelchair access required"
}

```

VisitorRequest:

3.3 Registrations - /api/v1/registrations

Method	Path	Description	Status Codes
GET	/registrations	List (filter by `?timeSlotId=`, `?date=`, or `?prospect=true`)	200
GET	/registrations/{id}	Get registration by ID	200, 404
POST	/registrations	Create registration (visitor->slot, requires `visitPurpose`)	201, 400, 409, 422
PATCH	/registrations/{id}/approve	Approve -> triggers briefing creation	200, 404, 422
PATCH	/registrations/{id}/reject	Reject a pending registration	200, 404, 422
PATCH	/registrations/{id}/check-in	Check in a confirmed visitor	200, 404, 422
PATCH	/registrations/{id}/cancel	Cancel a registration	200, 404, 422

```

{
  "visitorId": 1,
  "timeSlotId": 10,
  "visitPurpose": "PURCHASE",
  "eventId": null
}

```

RegistrationRequest:

```

{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "groupSize": 2,
  "timeSlotId": 10,
  "slotDate": "2026-09-08",
  "slotStart": "09:00",
  "eventId": null,
  "visitPurpose": "PURCHASE",
  "prospect": true,
  "status": "PENDING",
  "briefingId": null,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}

```

RegistrationResponse (new fields):

3.4 Briefings - /api/v1

Method	Path	Description	Status Codes
GET	/registrations/{registrationId}/briefing	Get briefing for registration	200, 404
PATCH	/registrations/{registrationId}/briefing/deliver	Mark briefing delivered	200, 422

```

{
  "staffMember": "Guard Camille",
  "signature": "digital-signature-or-name"
}

```

BriefingDeliverRequest:

3.5 Events - /api/v1/events

Method	Path	Description	Status Codes
GET	/events	List events (`?type=`, `?date=YYYY-MM-DD`)	200
GET	/events/{id}	Get event by ID	200, 404
POST	/events	Create event (default status `DRAFT`)	201, 400, 422

```
| `PUT` | `/events/{id}` | Update event (not allowed when CANCELLED/COMPLETED) | 200, 404, 422 |
| `DELETE` | `/events/{id}` | Cancel event (soft-delete) | 204, 404, 422 |
| `POST` | `/events/{id}/publish` | Publish a DRAFT event | 200, 404, 422 |
| `GET` | `/events/{id}/registrations` | List registrations for the event | 200, 404 |
| `POST` | `/events/{id}/register` | Register a visitor for the event | 201, 400, 404, 409, 422 |
```

```
{
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall"
}
```

EventRequest:

```
{
  "id": 1,
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "booked": 3,
  "status": "PUBLISHED",
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall",
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

EventResponse:

```
{
  "visitorId": 1,
  "visitPurpose": "PURCHASE"
}
```

EventRegistrationRequest:

The registration created for an event has timeSlotId: null and carries the visitor's visitPurpose. Commercial purposes (PURCHASE, PARTNERSHIP, INVESTMENT) automatically mark the visitor as a prospect, exactly like time-slot registrations.

3.6 Feedback & Surveys - /api/v1

```
| Method | Path | Description | Status Codes |
|-----|-----|-----|-----|
| `GET` | `/feedback` | List responses (`?visitorId=`, `?from=`, `?to=`) | 200 |
| `GET` | `/feedback/summary` | Compiled stats (`?from=`, `?to=`) | 200 |
| `POST` | `/feedback` | Submit response (on-site or linked to a sent survey) | 201, 400, 404, 422 |
| `PATCH` | `/feedback/{id}/route` | Route a response to a module owner | 200, 404, 400 |
| `GET` | `/surveys` | List sent surveys (`?visitorId=`, `?status=`) | 200 |
| `POST` | `/surveys` | Send a post-visit survey (email/SMS/WhatsApp) | 201, 400, 404, 422 |
```

```
{
  "visitorId": 1,
  "surveyId": 12,
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure"
}
```

FeedbackRequest: visitorId (required), surveyId (optional - links the response to a sent survey and marks it RECEIVED), rating (1-5, required), briefingClear, educationalValue, recommend, comment.

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "surveyId": 12,
  "origin": "EMAIL",
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure",
  "routedTo": "Sales (Mariata)",
  "submittedAt": "2026-08-26T13:00:00Z",
  "createdAt": "2026-08-26T13:00:00Z",
  "updatedAt": "2026-08-26T13:00:00Z"
}
```

FeedbackResponse:

```
{
  "averageRating": 4.2,
  "total": 18,
  "recommendPct": 92.0,
  "distribution": { "1": 0, "2": 0, "3": 3, "4": 6, "5": 9 }
}
```

FeedbackSummaryResponse (compiled weekly report):

```
{
  "visitorId": 3,
  "channel": "EMAIL",
  "messageTemplate": "Thank you for visiting Sustainable Farm! Share your feedback: [link]"
}
```

SurveyRequest:

SurveyResponse: id, visitorId, visitorName, channel, messageTemplate, sentAt, status (SENT/RECEIVED), rating (once the visitor responded), createdAt, updatedAt.

On-site tablet submissions use POST /feedback without surveyId -> origin: "ON_SITE". A sent survey becomes RECEIVED automatically when a response is linked to it.

3.7 Educational Program - /api/v1

Method	Path	Description	Status Codes
GET	/tour-stops	List guided-tour stops (ordered by position)	200
POST	/tour-stops	Create a tour stop	201, 400, 409
PUT	/tour-stops/{id}	Update a stop (position must stay unique among active stops)	200, 404, 409, 400
DELETE	/tour-stops/{id}	Deactivate a stop (soft-delete)	204, 404
GET	/workshops	List workshops/templates (?status=DRAFT ACTIVE INACTIVE)	200
POST	/workshops	Create a workshop (status DRAFT)	201, 400
PUT	/workshops/{id}	Update a workshop	200, 404, 400
POST	/workshops/{id}/publish	Publish a DRAFT workshop -> ACTIVE	200, 404, 422
POST	/workshops/{id}/deactivate	Deactivate an ACTIVE workshop -> INACTIVE	200, 404, 422

```
{
  "name": "Solar plant & tracking system",
  "position": 4,
  "description": "86.4 kwp installed, +20% yield with tracking",
  "durationMinutes": 15,
  "maxCapacity": 15,
  "location": "Solar field - observation path",
  "demo": "Live dashboard screen (Infineon sensors)",
}
```



```

"safetyNotes": "Observation path only; tracker, battery/inverter and running generator restricted"
}

```

TourStopRequest / TourStopResponse: name (required), position (required, ?1, unique among active stops), description, durationMinutes (required, ?1), maxCapacity (optional, ?1), location, demo, safetyNotes, active.

WorkshopRequest / WorkshopResponse: name (required), durationMinutes (required, ?1), targetGroup (required, e.g. All, Professional, Schools), facilitator, description, status (DRAFT by default).

Confirmed reference content (module owners: Aida, Jean-Louis, Delwende, Phares, Abdoul): standard tour stops - Welcome & safety briefing (5-10 min, Front Desk), Mango orchard (20 min, Keitt 200 trees/2 ha), Smart drip irrigation (10-15 min, max 10), Solar plant & tracking (15 min, max 15), Processing unit (20 min, max 10), Wrap-up & questions (15 min) -> max 10 pers/slot (limiting stops: irrigation + processing). Reference workshops: Standard farm tour (~100 min, All, Alix), Solar workshop (45 min, Professional, Guest Aida), Mango tasting & processing (40 min, General public, Guest Abdoul), School discovery day (120 min, Schools, P. Nikiéma).

3.8 Agritourism Booking System - /api/v1

Method	Path	Description	Status Codes
GET	/activities	List agritourism activities (alphabetical)	200
POST	/activities	Create an activity (name, price FCFA, capacity, durationMinutes)	201, 400
PUT	/activities/{id}	Update an activity	200, 404, 400
DELETE	/activities/{id}	Deactivate an activity (soft-delete)	204, 404
GET	/bookings	List bookings (?status=PENDING CONFIRMED COMPLETED CANCELLED, ?activityId=, ?date=)	200
GET	/bookings/occupancy	Occupancy % per activity	200
POST	/bookings	Create a booking (validates capacity)	201, 400, 404, 409
PUT	/bookings/{id}	Update a booking (PENDING/CONFIRMED only)	200, 404, 409, 422
POST	/bookings/{id}/pay	Record payment -> PAID	200, 404, 422
POST	/bookings/{id}/confirm	Confirm (only PENDING + PAID), sends confirmation email, schedules 24h reminder	200, 404, 422
POST	/bookings/{id}/complete	Mark a CONFIRMED booking COMPLETED	200, 404, 422
POST	/bookings/{id}/cancel	Cancel PENDING/CONFIRMED; paid -> REFUNDED; frees capacity	200, 404, 422

```

{
  "name": "Solar workshop",
  "price": 5000,
  "capacity": 15,
  "durationMinutes": 45,
  "description": "Guided visit of the 86.4 kwp solar plant with tracking"
}

```

AgriActivityRequest / AgriActivityResponse: name (150), price (BigDecimal ? 0, FCFA), capacity (? 1), durationMinutes (? 1), description (1000), active.

```

{
  "activityId": 1,
  "timeSlotId": 2,
  "visitorFullName": "Lucas Weber",
  "visitorEmail": "lucas@example.com",
  "visitorPhone": "+22612345678",
  "peopleCount": 5,
  "paymentMethod": "ORANGE_MONEY"
}

```

BookingRequest / BookingResponse: activityId, timeSlotId, visitorFullName, visitorEmail (valid email), visitorPhone, peopleCount (? 1), paymentMethod (CASH, ORANGE_MONEY, MOOV_MONEY). Response adds reference (BK-00001 format), activityName, activityPrice, slot date/times, totalAmount (price × people), paymentStatus, status, reminderScheduledAt.

Reference data from the mockup: Standard tour 5,000 FCFA/person (family rate 3,500 FCFA); payment accepted in cash, Orange Money, Moov Money.

Emails (confirmation + rappel 24 h). À la confirmation, un email de confirmation est envoyé au visiteur (référence BK-xxxxx, activité, date/créneau, personnes, total) et confirmationSentAt est posé. Un job planifié (BookingReminderJob, toutes les 60 s par défaut) envoie le rappel « moins de 24 h » aux réservations confirmées dont l'échéance est atteinte (reminderScheduledAt ? now) et passe reminderSentAt. Un échec d'envoi ne bloque jamais la réservation : l'erreur est loggée et la confirmation est maintenue. Deux implémentations de NotificationService :

Mode	Condition	Comportement
ConsoleNotificationServiceImpl	app.mail.enabled=false (défaut)	Email loggé en console (dev)
SmtplibNotificationServiceImpl	app.mail.enabled=true	Envoi réel via JavaMailSender (SMTP)

4. Business Rules

Scheduling Rules (SchedulingRules.java)

Rule	Value	Source
Max visitors per slot	**10**	Confirmed with module owners (Abdoul)
Max slots per day	**2**	Mockup: morning + afternoon
Closed day	**Sunday**	All module owners
Morning slot	09:00-11:00	Default
Afternoon slot	14:00-16:00	Default

Registration Rules

Rule	Enforcement
Cannot register on a **CANCELLED** slot	BusinessRuleException
No duplicate registration (visitor + slot)	ConflictException (409)
Group size must fit remaining capacity	BusinessRuleException
Only **PENDING** -> approve/reject	BusinessRuleException
Only **CONFIRMED** -> check-in	BusinessRuleException
Cannot cancel after **CHECKED_IN**	BusinessRuleException
CONFIRMED -> auto-creates Briefing (PENDING)	Service layer
Briefing delivery requires PENDING status	BusinessRuleException

Event Rules

Rule	Enforcement
endDateTime must be after startDateTime	BusinessRuleException
Only **DRAFT** events can be published	BusinessRuleException
Cannot publish an event that has already ended	BusinessRuleException
Cannot update **CANCELLED** or **COMPLETED** events	BusinessRuleException
Cannot cancel a **COMPLETED** event	BusinessRuleException
Registration only on **PUBLISHED** events	BusinessRuleException
No duplicate registration (visitor + event)	ConflictException (409)
booked + groupSize > maxCapacity -> reject	BusinessRuleException (422)
Booked count excludes REJECTED / CANCELLED registrations	Repository query

Feedback & Survey Rules

Rule	Enforcement
Rating must be between 1 and 5	Bean validation (400)
Response linked to a survey must match the survey's visitor	BusinessRuleException (422)
A survey can only be answered once (RECEIVED is final)	BusinessRuleException (422)
Survey channel cannot be ON_SITE (tablet responses are direct)	BusinessRuleException (422)
Response without a survey -> recorded as ON_SITE	Service layer
Linking a response marks the survey RECEIVED	Service layer
Routing targets are free-text tags (no coupling to other modules)	Data model

Educational Program Rules

Rule	Enforcement
-----	-----
Position must be ? 1	Bean validation (400)
Position must be unique among active stops	`ConflictException` (409)
Deactivating a stop frees its position (soft-delete)	`DELETE /tour-stops/{id}`
Create -> status `DRAFT`	Service layer
Only DRAFT workshops can be published	`BusinessRuleException` (422)
Only ACTIVE workshops can be deactivated	`BusinessRuleException` (422)

Agritourism Booking Rules

Rule	Enforcement
-----	-----
Capacity counts non-cancelled bookings only	Repository aggregate query
A booking can never exceed the slot capacity (10 pers)	`ConflictException` (409)
A booking can never exceed the activity capacity	`ConflictException` (409)
Bookings only on active activities and non-cancelled slots	`BusinessRuleException` (422)
Cannot update COMPLETED / CANCELLED bookings	`BusinessRuleException` (422)
Can only confirm a PENDING booking that is PAID	`BusinessRuleException` (422)
Confirm schedules the 24 h reminder (slot start ? 24h)	Service layer
Completing requires CONFIRMED status	`BusinessRuleException` (422)
Cancelling a PAID booking marks it REFUNDED	Service layer
Confirm sends the booking confirmation email	`NotificationService` (never blocks)
24 h reminder sent for due, unsent reminders	`BookingReminderJob` (fixed delay, 60 s)
Reminder / confirmation deliveries are idempotent (confirmationSentAt, reminderSentAt)	Repository query + service

Visitor Rules

Rule	Enforcement
-----	-----
Duplicate email ->	`ConflictException` (409) On create and update
@NotBlank on fullName	Bean validation (400)
@Email on email	Bean validation (400)

Lead Generation (Prospect Detection)

The visitor management app doubles as a marketing tool for the Sales & Marketing team. When a visitor registers with a commercial purpose, the system automatically flags them as a prospect.

How it works:

1. During registration, the visitor must select a visitPurpose (mandatory field)
2. If the purpose is PURCHASE, PARTNERSHIP, or INVESTMENT, the system sets isProspect = true automatically
3. Sales & Marketing can query all prospects via GET /api/v1/registrations?prospect=true

VisitPurpose values:

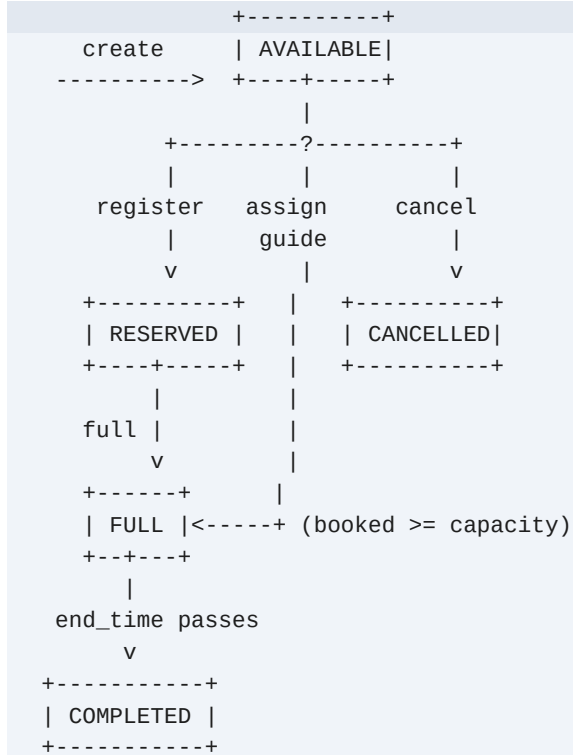
Purpose	Prospect?	Description
-----	-----	-----
`TOURISM`	No	Leisure visit, tourism
`PURCHASE`	Yes	Interested in buying products
`PARTNERSHIP`	Yes	Business partnership opportunity
`INVESTMENT`	Yes	Investment opportunity
`EDUCATION`	No	Training, educational visit
`OTHER`	No	Other reasons

GET /api/v1/registrations?prospect=true

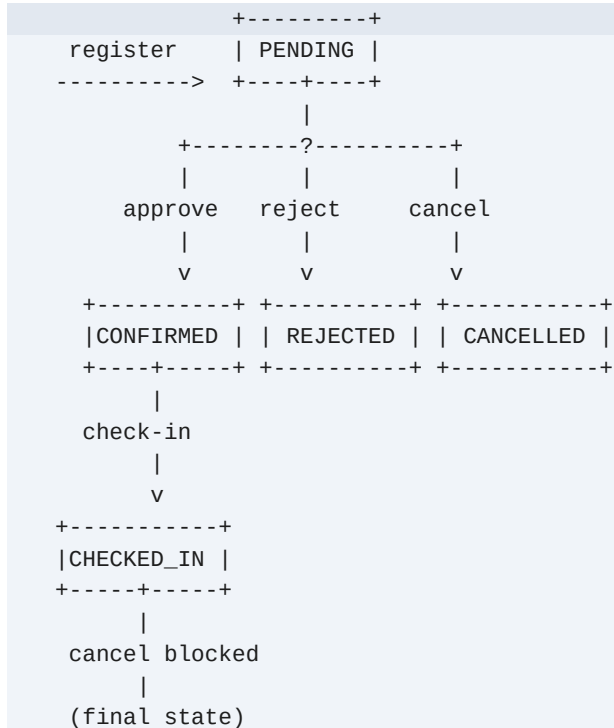
Prospect query example: Returns all registrations where isProspect = true, including visitor name, email, phone, and visit purpose.

5. Status Flow Diagrams

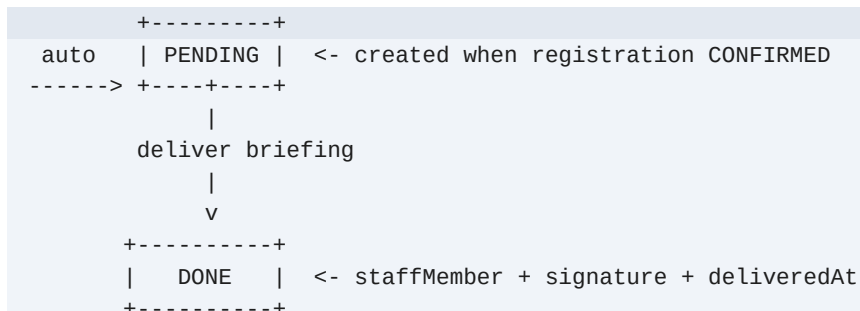
TimeSlot Status Flow



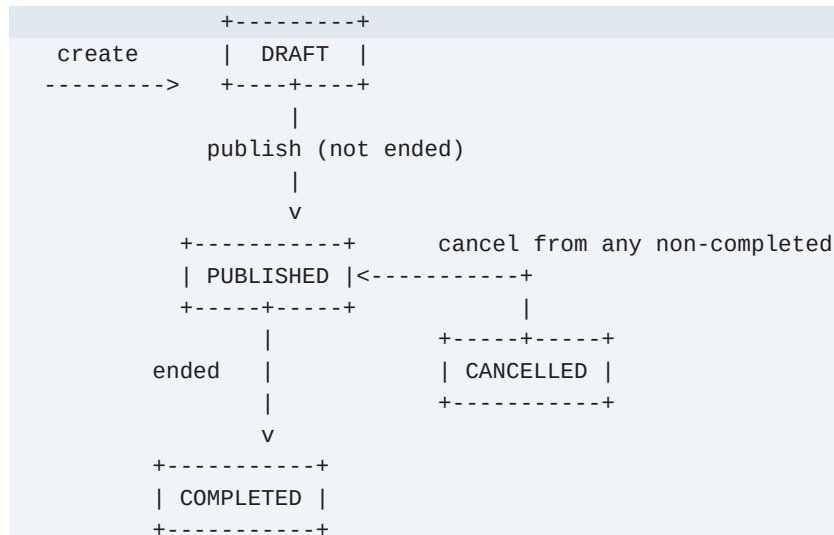
Registration Status Flow



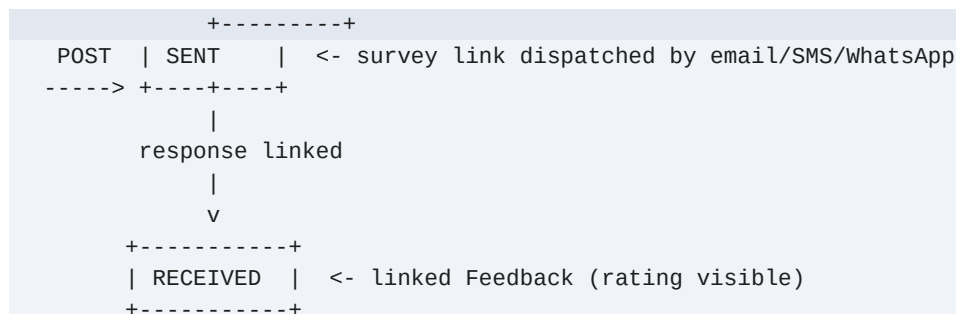
Briefing Status Flow



Event Status Flow



Survey Status Flow



6. Error Handling

All errors return a consistent JSON envelope via `GlobalExceptionHandler`:

```
{
  "status": 404,
  "error": "Not Found",
  "message": "Time slot 999 not found",
  "path": "/api/v1/time-slots/999"
}
```

Exception	HTTP Status	When
<code>ResourceNotFoundException</code>	404	Entity not found
<code>ConflictException</code>	409	Duplicate email, duplicate registration, overlapping slot
<code>BusinessRuleException</code>	422	Invalid state transition, capacity exceeded, Sunday creation
<code>MethodArgumentNotValidException</code>	400	Bean validation failures (<code>@NotBlank</code> , <code>@Email</code> , etc.)

7. Test Coverage Summary

Layer	Annotation	Tests	Framework
Repository	<code>@DataJpaTest</code>	39	Spring Data + AssertJ
Service	<code>@ExtendWith(MockitoExtension.class)</code>	94	Mockito + AssertJ
Controller	<code>@WebMvcTest</code>	82	MockMvc + Mockito
Context	<code>@SpringBootTest</code>	1	Spring Boot
Core (notification)	plain Mockito	3	Mockito + AssertJ
Total		**219**	

Test Files

```
src/test/java/com/infineonbit/sustainablefarm/
+-- SustainableFarmApplicationTests.java
+-- modules/visitormanagement/
    +-- repository/
        | +-- TimeSlotRepositoryTest.java      (5 tests)
        | +-- VisitorRepositoryTest.java       (4 tests)
        | +-- RegistrationRepositoryTest.java  (5 tests)
        | +-- BriefingRepositoryTest.java      (2 tests)
        | +-- EventRepositoryTest.java         (4 tests)
        | +-- FeedbackRepositoryTest.java      (7 tests)
        | +-- EducationalProgramRepositoryTest.java (5 tests)
        | +-- AgriTourismRepositoryTest.java (7 tests)
    +-- service/
        | +-- SchedulingServiceImplTest.java   (13 tests)
        | +-- RegistrationServiceImplTest.java (21 tests)
        | +-- EventServiceImplTest.java        (16 tests)
        | +-- FeedbackServiceImplTest.java     (12 tests)
        | +-- EducationalProgramServiceImplTest.java (12 tests)
        | +-- BookingServiceImplTest.java      (18 tests)
        | +-- BookingReminderJobTest.java      (2 tests)
    +-- controller/
        +-- TimeSlotControllerTest.java        (9 tests)
        +-- VisitorControllerTest.java         (7 tests)
        +-- RegistrationControllerTest.java     (12 tests)
        +-- BriefingControllerTest.java        (5 tests)
        +-- EventControllerTest.java           (13 tests)
        +-- FeedbackControllerTest.java        (10 tests)
        +-- EducationalProgramControllerTest.java (12 tests)
        +-- BookingControllerTest.java         (13 tests)
+-- core/notification/
    | +-- NotificationServiceTest.java         (3 tests)
```

8. Running the Backend

```
# Start PostgreSQL + Backend
docker compose up -d backend

# Backend available at
http://localhost:8080

# Swagger UI
http://localhost:8080/swagger-ui/index.html

# Actuator health
http://localhost:8080/actuator/health

# Run tests (Docker, no local JDK required)
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw test

# Compile only
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw compile
```

Email notifications (env vars)

Real SMTP delivery requires MAIL_ENABLED=true plus SMTP credentials. Without it, emails are only logged to the console (dev mode).

Variable	Default	Description
MAIL_ENABLED	false	true -> real SMTP, false -> console log

```
| `MAIL_HOST` | *(empty)* | SMTP host (e.g. `smtp.orange.bf`) |
| `MAIL_PORT` | `587` | SMTP port |
| `MAIL_USERNAME` | *(empty)* | SMTP username |
| `MAIL_PASSWORD` | *(empty)* | SMTP password |
| `MAIL_FROM` | `visits@sustainable-farm.local` | Sender address |
| `MAIL_REMINDER_INTERVAL_MS` | `60000` | Reminder poll interval (fixed delay) |
```

9. Technology Stack

Component	Version	Notes
Java	21	LTS
Spring Boot	4.1.0	Modular monolith
Spring Data JPA	4.1.0	Derived queries + `@Query`
Hibernate ORM	7.4.1	`ddl-auto=update` (no Flyway)
PostgreSQL	17-alpine	Production database
H2	2.x	Test-only in-memory DB
SpringDoc OpenAPI	2.8.5	Swagger UI auto-generated
Spring Mail	4.1.0	`JavaMailSender` + SMTP (emails booking)
Lombok	1.18.46	Boilerplate reduction
Jackson	3.1.4	JSON serialization
Mockito	5.23.0	Service + controller tests
AssertJ	3.27.7	Fluent assertions

Last updated: 2026-09-09